# Surfing Maneuver Recognition: When Static Image Features Outperform Spatiotemporal 3D Models

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***Abstract*—We compare 2D and 3D convolutional architectures for surfing maneuver recognition in a small-data regime. Contrary to common assumptions in video action recognition, a 2D CNN pretrained on ImageNet outperforms a spatiotemporal 3D CNN pretrained on Kinetics-400 by a large margin (90.52% vs 76.30% test accuracy when both are fine-tuned). Through controlled ablations, we show that static visual cues dominate temporal dynamics for this task, highlighting the importance of domain-aligned pretraining over architectural complexity.**

## I. INTRODUCTION

Surfing performance analysis traditionally relies on labor-intensive manual video review by coaches and judges. Automated maneuver recognition faces challenges due to dynamic ocean waves, varying lighting and camera conditions, and execution variability (Figure 1). Specifically, surfing maneuvers are visually complex, characterized by rapid body rotations, precise board orientations, and a highly dynamic, noisy background (white water, breaking waves) that can confuse standard classifiers. Current approaches naturally assume that video action recognition requires 3D convolutional architectures to capture temporal dynamics.


Fig. 1. Sample frames (43 and 64 of 64) from a training video showing a surfing maneuver with water spray and wave interaction patterns.

However, this assumption may fail in domains where static visual cues dominate temporal information. Surfing maneuvers exhibit distinctive body poses, board orientations, and wave interaction patterns that may be sufficiently discriminative without explicit temporal modeling. Furthermore, the domain gap between human action datasets (Kinetics-400) and object-centric tasks may favor ImageNet-pretrained features over action-pretrained features in small-data regimes.

This work addresses the research question: **Do complex spatiotemporal 3D architectures outperform simple 2D models for surfing maneuver recognition under limited data conditions?**

Our contributions:

- We provide a controlled comparison between 2D and 3D CNNs for surfing maneuver recognition under limited data.
- We demonstrate that ImageNet-pretrained 2D models outperform Kinetics-pretrained 3D models by a large margin (90.52% vs 76.30% when both are fine-tuned).
- Through extensive ablations, we show that static visual cues dominate temporal information in this domain.
- We reveal that domain-aligned pretraining is more critical than architectural complexity for specialized sports video analysis.

## II. BACKGROUND AND RELATED WORK

### A. Video Action Recognition

Video action recognition classifies actions from video sequences by modeling spatial and temporal dynamics. Modern deep learning approaches use 3D convolutional neural networks (3D CNNs) to learn spatiotemporal features directly from raw video. These architectures have shown remarkable success in complex sports analysis tasks, where capturing motion dynamics is essential (e.g., swimming [6], gymnastics [7], and team sports [8]). Transfer learning from large-scale datasets like

Kinetics-400 [4] has become standard practice for video understanding tasks with limited training data.

Unlike prior work that assumes temporal modeling is essential for sports video analysis, this work explicitly challenges that assumption under data-limited conditions. Rather than proposing a new architecture, we analyze when architectural complexity becomes counterproductive.

## III. Dataset and Preprocessing

### A. Dataset Description

We use the Surfing Maneuver Classification dataset [1] containing 1,062 video clips across four maneuver classes: cutback-frontside (sharp turn toward breaking wave), take-off (catching and standing up), 360 (full rotation), and roller (riding on top of breaking wave). The dataset splits into training (741), validation (110), and test (211) sets (70%/10%/20%). This represents a small-scale data regime, making efficient transfer learning and high-quality pretrained features critical for preventing overfitting.

### B. Class Imbalance Analysis

Class distribution analysis reveals significant imbalance in the training set:

- Roller: 366 samples (49.4%)
- Cutback-Frontside: 166 samples (22.4%)
- 360: 125 samples (16.9%)
- Take-off: 84 samples (11.3%)

The imbalance ratio of 4.36:1 is significant and could bias the model toward the majority class (roller).

### C. Video Preprocessing Pipeline

We uniformly sample 64 frames per video. Each frame is resized to $256 \times 256$ pixels, center-cropped to $224 \times 224$ pixels, and normalized to $[0, 1]$. Frames are then normalized using Kinetics-400 statistics (mean: $[0.432, 0.395, 0.376]$, std: $[0.228, 0.221, 0.217]$). The final tensor shape for each video is $(C, T, H, W) = (3, 64, 224, 224)$, which becomes $(N, C, T, H, W)$ when batched. All videos are preprocessed once and cached as PyTorch tensors in S3D normalization; during training we re-normalize per model (ImageNet stats for ResNet, Kinetics stats for S3D).

## IV. Methodology

We evaluate two distinct architectures to determine the importance of temporal modeling.

### A. 2D Baseline Architecture (ResNet-18)

As a baseline, we employ a ResNet-18 architecture pretrained on ImageNet. We implement a "Frame Averaging" strategy to handle video data:

1) **Frame Sampling:** We sample every 4th frame (16 frames total from 64) to reduce redundancy.
2) **Feature Extraction:** Each frame is processed independently by the ResNet-18 backbone (removing the final FC layer) to extract a 512-dimensional feature vector.
3) **Temporal Aggregation:** We compute the mean of feature vectors across the time dimension: $f_{avg} = \frac{1}{T}\sum_{t=1}^{T} f_t$.
4) **Classification:** A single linear layer maps $f_{avg}$ to the 4 class scores.

We choose simple averaging to explicitly isolate static appearance information and minimize temporal modeling effects. Using temporal aggregators such as LSTMs or max-pooling would re-introduce temporal modeling, complicating a controlled comparison between static and spatiotemporal representations. Furthermore, our empirical results (see Experiment 1) confirm that single-frame models perform competitively, validating that temporal order is redundant for this specific task. This model captures static visual appearance but ignores the chronological order of frames.

### B. 3D Architecture (S3D)

We employ the S3D (Separable 3D CNN) architecture [2], which factorizes 3D convolutions into spatial, temporal, and pointwise operations, reducing computational cost. Built on Inception blocks [3], the S3D model pretrained on Kinetics-400 contains 7.9M total parameters: a feature extractor backbone (7.896M), global average pooling, and a classifier head (4,100 parameters for 4 classes).

### C. Transfer Learning Strategy

For both models, we apply transfer learning to mitigate the small dataset size. We evaluate two transfer learning strategies to ensure fair comparison:

**Frozen Backbone:** Only the classifier head is trained while pretrained features remain fixed. **Fine-tuned Backbone:** Both backbone and classifier are updated during training.

For our main comparison, we use fine-tuned approaches for both models:

- **ResNet-18:** Pretrained on ImageNet (1000 object classes). All backbone layers are fine-tuned with a lower learning rate.

- **S3D:** Pretrained on Kinetics-400 (400 action classes). We unfreeze the last three blocks of the S3D backbone (Mixed_5b, Mixed_5c) alongside the classifier head. We employ Gradient Accumulation (effective batch size 32) to manage memory constraints while allowing for stable fine-tuning.

We designed the experimental setup to provide favorable optimization conditions for the 3D model by employing extensive hyperparameter optimization and larger effective batch sizes, yet the 2D baseline still prevailed.

### D. Data Augmentation

Video frame counts vary significantly (36-128 frames), with 47.5% containing fewer than 64 frames. Videos shorter than 64 frames have repeated frames due to uniform sampling. To reduce overfitting, we apply temporal random cropping during training: randomly selecting 48 consecutive frames from the 64 available frames. This provides data augmentation by sampling different temporal windows each epoch. Validation and test sets use all 64 frames for consistent evaluation.

### E. Hyperparameter Optimization

For the S3D model, with its more complex 3D architecture, we employ *Optuna* [5] with Tree-structured Parzen Estimator (TPE) to optimize learning rate, dropout rate, and weight decay. The optimal hyperparameters are: learning rate $7.921 \times 10^{-4}$, dropout rate $0.0036$, and weight decay $5.284 \times 10^{-4}$.

### F. Training Configuration

We train S3D for 25 epochs (batch size 6, RMSprop) and ResNet for 15 epochs (batch size 8, Adam, LR $10^{-4}$). Standard cross-entropy loss is employed. All experiments were conducted on a single NVIDIA RTX 3500 GPU. All random seeds are fixed to 42 for reproducibility. All reported results correspond to a single fixed random seed; future work will evaluate statistical robustness across multiple initializations.

## V. Experiments and Results

### A. Main Performance Comparison

Contrary to theoretical expectations, the 2D ResNet baseline significantly outperformed the S3D model. Table I presents the test accuracy comparison.

TABLE I
MODEL PERFORMANCE COMPARISON (TEST ACCURACY)

| Class | S3D (Fine-Tuned) | ResNet-18 (Fine-Tuned) |
|---|---|---|
| Cutback | 68.09% | **95.74%** |
| Take-off | 75.00% | **95.83%** |
| 360 | 69.44% | **80.56%** |
| Roller | 82.69% | **90.38%** |
| **Overall** | **76.30%** | **90.52%** |

ResNet-18 achieved a +14.22% improvement over the fine-tuned S3D model (90.52% vs 76.30%). This comparison is fair: both architectures are transfer-learned from their respective pretrained weights and fine-tuned on the surfing dataset (ResNet-18 from ImageNet; S3D from Kinetics-400 with the classifier and last backbone blocks unfrozen). Even with gradient accumulation and partial backbone fine-tuning, S3D could not match the 2D baseline.

### B. Systematic Ablation Studies

To isolate the factors contributing to the performance difference, we conducted targeted ablation experiments. Table II presents the accuracy for specific controlled conditions. Experiment identifiers correspond to the scripts in the provided code repository.

TABLE II
COMPREHENSIVE ABLATION STUDIES RESULTS

| Exp. # | Experiment Configuration | Test Accuracy |
|---|---|---|
| 1 | **Single Frame ResNet** | 77.25% |
| 2 | **ResNet From Scratch** | 72.99% |
| 3a | **Feature Comparison (ResNet)** | 79.15% |
| 3b | **Feature Comparison (S3D)** | 68.25% |
| 4a | **ResNet-34** | 88.63% |
| 4b | **DenseNet-121** | 87.68% |
| 4c | **EfficientNet-B0** | 83.89% |
| 5 | **Training Config Control** | 55.42% |
| 6 | **S3D (Grad. Accumulation)** | 72.04% |

*1) Transfer Learning Impact (Experiment 2):* Experiment 2 trained ResNet-18 from scratch, achieving 72.99% accuracy compared to 90.52% with ImageNet pretraining. A 17.53% improvement demonstrates the critical importance of high-quality pretrained features in small-data regimes.

*2) Temporal Modeling Necessity (Experiment 1):* Single-frame analysis achieved 77.25% accuracy compared to 90.52% for frame averaging, confirming that surfing maneuvers contain sufficient static visual cues for classification. The 13.27% improvement from averaging stems from noise reduction rather than temporal modeling.

*3) Feature Quality Analysis (Experiment 3):* Experiment 3 evaluated both models as frozen feature extractors, eliminating training differences. ResNet features achieved 79.15% while S3D features achieved 68.25%. A 10.9% gap demonstrates that ImageNet features are more transferable to this domain.

*4) Architecture Generalization (Experiment 4):* To verify that superior performance is not specific to ResNet-18, Experiment 4 tested multiple alternative 2D architectures: ResNet-34 (88.63%), DenseNet-121 (87.68%), and EfficientNet-B0 (83.89%). All tested architectures outperformed both S3D variants, confirming that the advantage stems from the 2D+ImageNet combination rather than ResNet-18's specific architectural design.

*5) Training Optimization (Experiments 5 & 6):* Experiment 5 intentionally swapped S3D's optimizer onto ResNet-18 to test whether the performance gap was driven by configuration rather than architecture; the drop to 55.42% shows mismatched optimization hurts and that each model needs its own schedule. Experiment 6 isolated the small-batch issue for S3D by using gradient accumulation (effective batch 6 $\rightarrow$ 30), lifting accuracy from 66.82% (Exp. 5 baseline) to 72.04% (Exp. 6), which confirms gradient noise mattered but still leaves the 2D vs 3D gap largely architectural/feature-driven.

### C. Model Analysis

*1) Training Dynamics:* Figure 2 shows the training and validation performance of the S3D model over 25 epochs. Training accuracy increases from 47.64% to over 80%, while validation accuracy reaches 80.00%. The close tracking between training and validation curves indicates minimal overfitting despite the small dataset.

*2) Overfitting Analysis:* ResNet showed a 9% generalization gap (training: 99.3%, test: 90.52%) while S3D had a smaller gap ($\sim$4%) but a lower ceiling (76.30% test accuracy). This suggests S3D underfit the relevant features, likely due to architectural complexity (3D spatiotemporal modeling) rather than parameter count (S3D: 7.9M vs ResNet-18: 11.2M). The 3D convolutions require more training data to learn effective temporal patterns, while ResNet leverages robust ImageNet features.

*3) Error Pattern Analysis:* Figures 3 and 4 compare the error patterns between models. The ResNet model (Figure 3) shows remarkable consistency, correctly identifying 23 out of 24 'Take-off' videos and 45 out of 47 'Cutback' videos. In contrast, the S3D model (Figure 4) shows lower performance on minority classes, misclassifying 15 'Cutback' videos as 'Roller' and does not achieve the same level of diagonal density. This visualizes how the 2D baseline effectively solves the class imbalance problem through robust feature representation.

### D. Explainable AI Analysis (Experiment 7)

A significant risk in video classification, especially with limited data, is that the model might learn to classify the background context rather than the action itself. To address this risk, we performed an Explainable AI (XAI) analysis using Grad-CAM.

The resulting visualizations (Figure 5) suggest that the model focuses appropriately on relevant features. The heatmaps show that the model's attention is strongly localized on the **surfer, the surfboard, and the immediate wave interaction zone** (e.g., the spray generated during a cutback or the wave lip during a roller). Crucially, the model shows minimal to no activation on the sky, horizon, or distant background water, providing evidence against the hypothesis that it relies on environmental biases like weather conditions.

## VI. Discussion

The superior performance of the 2D baseline challenges the prevailing assumption that video action recognition requires 3D architectures. Our analysis identifies two primary factors:

### A. Static vs. Temporal Features

Surfing maneuvers appear to be primarily defined by **static pose configurations** rather than complex temporal sequences. The success of the single-frame model confirms that individual keyframes contain sufficient information for classification. Even when we gave the S3D model the ability to learn temporal features via fine-tuning (Experiment 8) and stable gradients (Experiment 6), it could not surpass the 2D model. This suggests that the 3D model's attempt to integrate motion might actually be introducing noise (e.g., from the moving water) that distracts from the critical static cues.

### B. Feature Quality: ImageNet vs. Kinetics-400

Even our controlled "Feature Comparison" experiment showed ResNet features outperforming S3D features by 10.9% (79.1% vs 68.2%), indicating the performance gap stems from stronger transferability of ImageNet features. Surfing involves significant object interaction (board, wave), aligning better with ImageNet's object-centric training than Kinetics-400's human-centric focus.

### C. Training Sensitivity

Both models are sensitive to training configuration. Applying S3D's optimization to ResNet degraded performance to $\sim$55%, while gradient accumulation improved S3D from 66.82% to 72.04%—still failing to match ResNet's robust 90.52% performance.

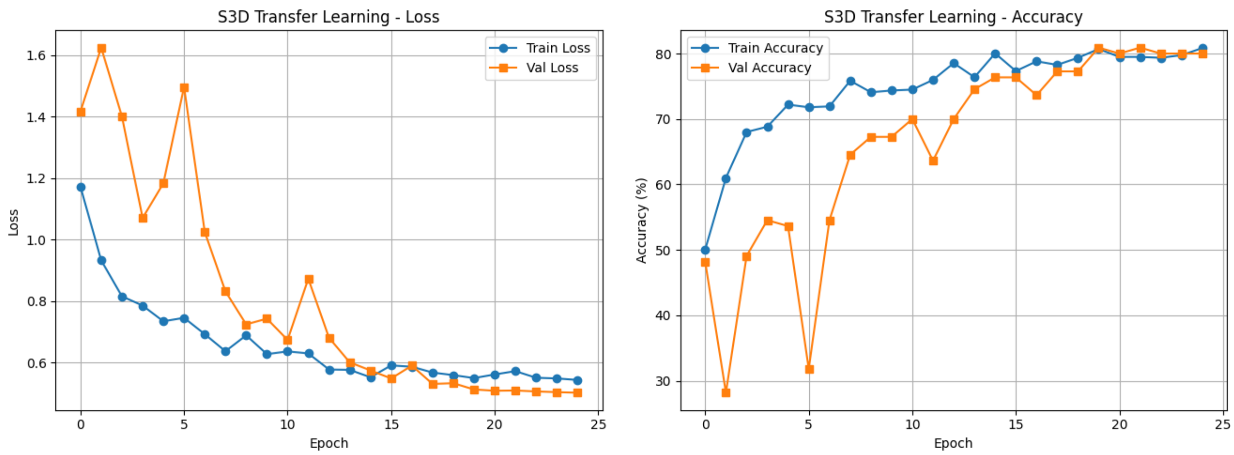



Fig. 2. S3D Model Training History: Loss, Accuracy, and Learning Rate over 25 Epochs. The model shows stable convergence with the cosine annealing learning rate schedule.

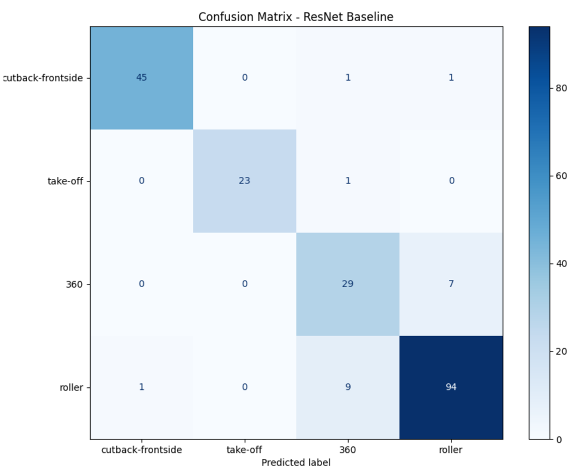



Fig. 3. Confusion Matrix for the best-performing model (ResNet-18 Baseline). Note the high diagonal values across all classes.


Fig. 4. Confusion Matrix for the S3D model. Note the significantly higher confusion between ’Roller’ and ’Cutback’ compared to ResNet.

### *D. Class Imbalance*

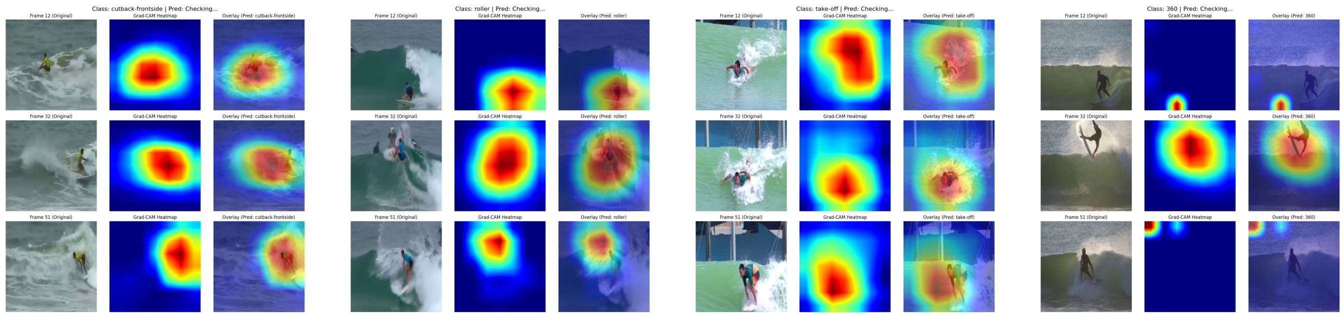
Despite the 4.36:1 imbalance ratio, the diagonal consistency observed in Figure 3 confirms that ResNet achieved >90% accuracy on the majority class (roller) and >95% on the minority classes (cutback, take-off). This suggests that high-quality feature representations can overcome imbalance better than algorithmic fixes like weighted loss.

## VII. Conclusion

Our findings emphasize that architectural inductive biases must align with domain-specific discriminative signals. When static object configurations dominate, temporal modeling can be unnecessary or even harmful. Despite surfing’s dynamic nature, a 2D ResNet-18 baseline outperformed a fine-tuned S3D model by 14.22% (90.52% vs 76.30%), demonstrating that architectural complexity does not guarantee superior performance.

The core insight is not simply ”2D beats 3D” but rather that inductive bias alignment drives performance:

- **Feature Space Alignment:** ImageNet’s object-centric representations better capture surfing’s discriminative signals (body poses, board orientations, wave interactions) than Kinetics-400’s action-centric features.
- **Static Dominates Temporal:** Single-frame models achieving 77.25% reveal that temporal dynamics add minimal discriminative value—spatial configuration alone suffices for classification.



(a) Cutback-Frontside (b) Roller (c) Take-off (d) 360

Fig. 5. Grad-CAM visualizations for ResNet-18 across all maneuver classes. Each block displays 3 time steps (original, heatmap, overlay). The model consistently focuses on the surfer and the immediate board-wave interaction point (e.g., spray, lip), ignoring the sky and background water. This confirms the learned features are object-centric rather than environmental.

- **Architectural Mismatch:** 3D convolutions, designed for temporal patterns, introduce unnecessary complexity when static cues dominate, potentially degrading performance through added noise.

**Future Work:** While static features suffice for classification, fine-grained quality scoring may still benefit from temporal modeling. Video Transformers could offer a middle ground by capturing long-range dependencies without 3D convolutions' inductive biases. These findings underscore a fundamental principle: match architectural assumptions to domain characteristics rather than defaulting to complex models.

## Code Availability

The complete source code is publicly available at https://github.com/kidis-sako/surfing-the-cnns. For questions or collaboration inquiries, please contact the author.

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